

# The Self-Consistent $D$ -Composite Scheme at Linear Order: Sector Constraint, Symbol Identity, Off-Resonance Flatness, and the Six-Dimensional Resonant Sector

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## Abstract

We begin the audit of the last unanalyzed variational scheme of the minimal branch: the self-consistent  $D$ -composite system in which the connection sits inside the defining jet. Four exact results are proved at the linearized, frozen-coefficient level around the affine point. (i) The Lorentz-real sector requires  $\Theta \in W_L$  identically; the Hermitian freedom of the Palatini analysis does not exist here. (ii) The frozen symbol  $A(s, \lambda)$  of the linearized implicit system satisfies the exact operator identity  $A^3 = (\lambda \cdot s) A^2$ ; its spectrum is  $\{0^{10}, q^6\}$  with  $q = \lambda \cdot s$ , with three nilpotent Jordan blocks on the kernel and  $\det(I - A) = (1 - q)^6$ . (iii) Off the single resonance hyperplane  $q = 1$  the unique  $\delta\Theta$ -driven solution is the exact gradient; the linearized tetrad is holonomic and the linearized geometry pullback-flat. (iv) At  $q = 1$  there is an exactly six-dimensional space of homogeneous resonant modes —  $\dim \mathfrak{so}(1, 3)$  of them — and every one is anholonomic with linearly independent curls. All linearized anholonomy, hence all candidate linearized curvature, is confined to this resonant sector. The variable-coefficient assembly of resonant modes, the quadratic action on the resonant sector, on-shell torsion, and the chain-rule spin current remain open. All algebraic statements are verified in `tools/verify_dcomposite_linearized.py`.

## 1 The self-consistent system

The scheme couples the covariant jet back to the connection:

$$E_\mu = \mathcal{N}_0^{-1/2} (\partial_\mu \Theta + \Omega_\mu[E] \Theta + \Theta \Omega_\mu[E]^\dagger), \quad g_{\mu\nu} = \frac{1}{2} (E_\mu^\dagger E_\nu + E_\nu^\dagger E_\mu), \quad (1)$$

with  $\Omega_\mu[E]$  the spin representation of the Levi–Civita connection of the tetrad defined by the slice coordinates of  $E_\mu$  (torsion-free branch first; anholonomy, not torsion, is the requirement for curvature).

**Lemma 1** (Sector constraint).  *$E_\mu \in W_L$  for all connection values forces the Hermitian part of  $\Theta$  to vanish identically: the Lorentz-real  $D$ -composite sector is  $\Theta(x) \in W_L$ . (Exact solve; check `L0`.)*

## 2 Linearization and the frozen symbol

The affine field  $\Theta_{\text{aff}} = \Theta_0 + \sqrt{\mathcal{N}_0} \bar{E}_\nu x^\nu$  with  $\Theta_0 \in W_L$  is an exact solution of (??) with  $\Omega = 0$ . Linearize  $\Theta = \Theta_{\text{aff}} + \varepsilon\theta$ ,  $E = \bar{E} + \varepsilon F$ , freeze coefficients at  $x_0$ , and pass to the plane-wave symbol  $\partial_\nu \rightarrow s_\nu$ . The system becomes

$$(I - A(s, \lambda)) F = \partial\theta', \quad \lambda^a = \text{slice coords of } \Theta_{\text{aff}}(x_0) / \sqrt{\mathcal{N}_0} = x_0^a + \theta_0^a, \quad (2)$$

where  $A$  implements the linearized Levi–Civita connection of the perturbed tetrad acting on the background through the bimodule.  $\mathcal{N}_0$  cancels;  $A$  is bilinear in  $(s, \lambda)$ . Under the diagonal Lorentz group at the identity-tetrad background the mixed contraction  $q = \lambda^a s_a$  is invariant.

**Theorem 1** (Symbol identity and spectrum).  $A^3 = q A^2$  holds as a polynomial identity in all eight variables (check D2, symbolic). Generically  $\text{rank } A = 9$ ,  $\text{rank } A^2 = 6$ ; hence the characteristic polynomial is  $t^{10}(t - q)^6$ , the minimal polynomial  $t^2(t - q)$ , and  $\det(I - A) = (1 - q)^6$  (check D3). The linearized system is uniquely solvable iff  $q \neq 1$ .

**Theorem 2** (Off-resonance flatness). The linearized Levi-Civita connection factors through the curl of the tetrad perturbation, so  $A$  annihilates every exact gradient (check D1). Consequently, for  $q \neq 1$  the unique  $\delta\Theta$ -driven solution of  $(I - A)F = \partial\theta'$  is  $F = \partial\theta'$  itself: exactly holonomic, with pullback-flat linearized geometry (check D4).

**Theorem 3** (Resonant sector). At  $q = 1$  the eigenspace  $\ker(I - A)$  is exactly six-dimensional ( $= \dim \mathfrak{so}(1, 3)$ ). Every resonant mode is anholonomic and the six curls are linearly independent (check D5, exact at a generic resonant point). All linearized anholonomy of the  $D$ -composite scheme is confined to this sector.

**Corollary 1** (Where linearized curvature can live). Since curvature at linear order requires anholonomy, any linearized gravitational content of the  $D$ -composite scheme must be assembled from resonant modes on the moving hyperplane  $(x_0 + \theta_0) \cdot s = 1$ . Off it, the scheme is flat at linear order — but, unlike the exact-gradient restriction, not identically: the resonant sector is a genuine nontrivial remainder.

### 3 Answers to the six audit questions at this level

(1) Local uniqueness of  $E[\Theta]$ : yes iff  $q \neq 1$ , with a single resonance hyperplane of multiplicity six. (2) Anholonomy: driven solutions are exactly holonomic; anholonomy exists only in the resonant sector. (3)  $R \neq 0$  without inserting curvature by hand: at linear order only through resonant modes; whether they assemble into curved solutions of the variable-coefficient system is open. (4) Chain-rule spin current: not yet computed. (5) On-shell torsion selection: not yet computed. (6) Degeneracy: off resonance the scheme degenerates to the flat sector, but not identically — the six-dimensional resonant remainder is the irreducible core on which the quadratic action must be evaluated next.

#### Scope and non-claims

Frozen-coefficient (microlocal) analysis at the affine point. Not a solution theory for the variable-coefficient system:  $\lambda$  moves with  $x_0$ , so the resonance set sweeps spacetime and the constant-coefficient solvability statements do not globalize automatically. Nothing here derives the Hilbert–Palatini term, decides on-shell torsion, or establishes curved solutions. A discovered-and-fixed index transposition in the linearized connection (which leaked the symmetric tetrad part into  $\omega^1$ ) is documented in the verifier as regression check D1; results computed with the faulty formula (an earlier  $\{0^4, q^9, (q/2)^3\}$  spectrum) are void.